

WEEK 6

Week 6A — Client-Server Using Named Pipes (FIFO)

This program demonstrates communication between a client and server using a **named pipe (FIFO)**.

The client sends a message through the FIFO, and the server reads and displays it. This covers the FIFO part of Question 6.

```
sri_dhruthi@Dhruthi: ~  
GNU nano 8.7.1 week6a_client  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
#include <string.h>  
  
int main()  
{  
    char message[] = "Hello from Client";  
  
    int fd = open("myfifo", O_WRONLY);  
  
    write(fd, message, strlen(message) + 1);  
  
    printf("Client sent: %s\n", message);  
  
    close(fd);  
  
    return 0;  
}
```

```
sri_dhruthi@Dhruthi: ~/OS/week6  
sri_dhruthi@Dhruthi:~/OS/week6$ gcc week6a_server.c -o week6a_server  
sri_dhruthi@Dhruthi:~/OS/week6$ ./week6a_server  
Server received: Hello from Client
```

```
sri_dhruthi@Dhruthi: ~/OS/week6  
sri_dhruthi@Dhruthi:~/OS/week6$ gcc week6a_client.c -o week6a_client  
sri_dhruthi@Dhruthi:~/OS/week6$ ./week6a_client  
Client sent: Hello from Client
```

Week 6B — POSIX Signal Handling

This program captures **SIGINT**, **SIGTERM**, and **SIGUSR1** using signal handlers. It demonstrates asynchronous event handling when signals are sent to the running process.

```
GNU nano 8.7.1 week6b.c
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handler(int sig)
{
    if (sig == SIGINT)
        printf("\nSIGINT Received\n");

    if (sig == SIGTERM)
        printf("\nSIGTERM Received\n");

    if (sig == SIGUSR1)
        printf("\nSIGUSR1 Received\n");
}

int main()
{
    signal(SIGINT, handler);
    signal(SIGTERM, handler);
    signal(SIGUSR1, handler);

    printf("Process ID: %d\n", getpid());

    while (1)
    {
        printf("Running...\n");
        sleep(3);
    }

    return 0;
}
```

```
sri_dhruthi@Dhruthi: ~/OS/week6
sri_dhruthi@Dhruthi:~/OS/week6$ gcc week6b.c -o week6b
sri_dhruthi@Dhruthi:~/OS/week6$ ./week6b
Process ID: 8192
Running...
Running...
Running...
Running...
SIGUSR1 Received
Running...
Running...
SIGTERM Received
Running...
^C
SIGINT Received
Running...
```

```
sri_dhruthi@Dhruthi: ~/OS/week6
sri_dhruthi@Dhruthi:~/OS/week6$ kill -SIGUSR1 8192
sri_dhruthi@Dhruthi:~/OS/week6$ kill -SIGTERM 8192
# You can also press Ctrl+C in the running terminal
sri_dhruthi@Dhruthi:~/OS/week6$
```

